

Natural class reasoning in segment deletion rules*

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1. Introduction

Logical Phonology (LP; e.g., Bale et al. 2014, Bale and Reiss 2018, Bale et al. 2020, Reiss 2021, Dabbous et al. 2024, Leduc et al. 2024, Reiss 2024, Gorman and Reiss 2025, 2026, Gorman 2026) is characterized by an insistence on a rigorous definition of the natural classes defining rule targets and environments, and by three novel operators which replace the $\rightarrow$ of classical generative phonology. Following these ideas to their logical conclusion has a counterintuitive consequence for the analysis of segment deletion rules:

(1) DELETE THE RICH: If some—but not all—X’s delete in some phonological context $Y \text{ — } Z$, the X’s that delete must be **more richly specified** than those which do not.

After some formal preliminaries (section 2), this prediction is illustrated using deletion processes in Hungarian (section 3) and Turkish (section 4). In Hungarian, there is independent evidence for underlyingly distinct sources of *h*, and these two *h*’s show differential behavior with respect to deletion. In Turkish, positing distinct sources for deleting and non-deleting *k* allows for a unified handling, within the narrow phonology, of apparent exceptions and an atypical form of non-derived environment blocking. These case studies show that DELETE THE RICH is not an empirical problem for LP.

2. Formal preliminaries

This section reviews the theory of rules in Logical Phonology. Some details not strictly relevant to deletion rules are set aside. Note that LP is a work in progress and the fragment presented here is not presented as a complete theory of phonology. Rather, LP is embedded in a network of related assumptions and theories. It is *substance-free* in that it rejects the notion of markedness in all its senses (e.g., Reiss 2017). Rather than deriving typological tendencies and gaps from markedness, LP takes these patterns to reflect an extra-grammatical

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diachronic filter—as developed in the works of Blevins (2004), Hale (2003, 2007), and Ohala (2003), among others—or from substantive properties of the language acquisition device (e.g., Gorman and Reiss 2025). It adopts Cognitive Phonetics (e.g., Volenec and Reiss 2018, 2019, 2026) as a theory of the universal phonology-phonetics interface, and thus rejects the hypothesis of language-specific phonetic implementation.

2.1 Features, segments, and natural classes

LP assumes that universal grammar provides a universal innate feature set (e.g., Chomsky and Halle 1965, Reiss and Volenec 2022), and further that these features are binary-valued (i.e., equipollent). A *feature specification* is thus a (possibly empty) set of valued features.

The feature specifications of segments must be *consistent*: if a segment is specified +F, it cannot also be specified −F and vice versa. However, feature specifications do not need to be *complete*: underspecification is permitted. Note that this should not be considered an axiom of LP; rather, completeness is a commonly-adopted axiom which LP happens to lack. In this sense LP is formally simpler than models forbidding underspecification (Reiss 2012).

Segments are consistent feature specifications linked to an X-slot. Some examples are given below; note that the $\square$ symbol in the specification of /I/ has no theoretical status, but is just intended to draw the reader’s attention to /I/’s underspecification with respect to HIGH.

(2) Sample segments:

$$\begin{array}{ccc} \begin{array}{c} \text{X} \\ | \\ /i/ = \left\{ \begin{array}{l} +\text{SYLLABIC} \\ -\text{BACK} \\ -\text{ROUND} \\ +\text{HIGH} \\ -\text{LOW} \\ +\text{ATR} \end{array} \right\} \end{array} & \begin{array}{c} \text{X} \\ | \\ /e/ = \left\{ \begin{array}{l} +\text{SYLLABIC} \\ -\text{BACK} \\ -\text{ROUND} \\ -\text{HIGH} \\ -\text{LOW} \\ +\text{ATR} \end{array} \right\} \end{array} & \begin{array}{c} \text{X} \\ | \\ /I/ = \left\{ \begin{array}{l} +\text{SYLLABIC} \\ -\text{BACK} \\ -\text{ROUND} \\ \square\text{HIGH} \\ -\text{LOW} \\ +\text{ATR} \end{array} \right\} \end{array} \end{array}$$

Natural classes are also defined by sets of valued features, but following Bale et al. (2020), who formalize a common intuition, they are interpreted as sets of segments. That is, a natural class “matches” (i.e., contains as a member) any segment whose specification is a superset of the natural class’s specification. For example, the following shows the intension and extension of the natural class defined by +HIGH.

$$(3) \ [+HIGH] = \{x \mid x \supseteq \{+HIGH\}\} = \{i, ptmi, y, \gamma, \dot{i}, \mathfrak{u}, \mathfrak{u}, u, \bar{u}, \dots\}$$

For type consistency, curly braces are used to denote segments (and sets in general) and square brackets are used for natural classes. As a notational convenience, a circle, along with a phoneme symbol and square brackets, is used to abbreviate natural classes.

$$(4) \ [\textcircled{i}] = \{x \mid x \supseteq \{+SYLLABIC, -BACK, -ROUND, +HIGH, -LOW, +ATR\}\} = \{i\}$$

LP assumes that targets and triggers (i.e., contexts or environments) of individual rules are intensionally defined by natural classes.¹ This does not entail that all phonological processes, informally speaking, have an extension which is itself a natural class. For example, a rule might extensionally target a set of segments that is smaller than its intensionally defined natural class because it is bled by an earlier rule removing potential targets; see Reiss and Volenec 2022:29f. for a concrete example involving English regular plurals. This does not exhaust the possible divergences between the intension and extension.

Of course, some natural classes, like (4), have a single-phoneme extension, and can thus serve as targets and triggers under this assumption. However, one consequence of this definition of natural classes—noted by Lees (1961:12–14) and Lightner (1971:236), among others—is that one cannot specify a natural class that contains an underspecified segment, say /I/, but does not also contain all segments whose specification is a proper superset of that segment, say /i, e/. This is exemplified and schematized below.

$$(5) [\textcircled{\text{I}}] = \{x \mid x \supseteq \{+\text{SYLLABIC}, -\text{BACK}, -\text{ROUND}, -\text{LOW}, +\text{ATR}\}\} = \{i, e, \text{I}\}$$

(6) SPECIFICITY: Let f, g be segments (feature specifications) such that $f \subset g$. Then there is no natural class containing f but excluding g .

As will be seen, DELETE THE RICH is a direct consequence of SPECIFICITY.

2.2 Rules

LP replaces the traditional $\rightarrow$ operator with three operators defined below. The *subtraction* operator $\setminus$, which corresponds to set difference, removes valued features from a segment.

$$(7) \text{ Subtraction: } A \setminus B = \{cF \mid cF \in A \wedge cF \notin B\}$$

The *unification* operator $\sqcup$, adapted from syntactic theory (e.g., Shieber 1986), adds a valued feature to a segment, but has no effect on targets already specified for that feature.

$$(8) \text{ Unification: } A \sqcup B = A \cup \{cF \mid cF \in B \wedge -cF \notin A\}$$

In other words, $A \sqcup B$ contains all features of A as well as any features of B which do not conflict with those in A . Thus unification only adds valued features when the target (the first argument) is underspecified with respect to the features in the change. Unification is thus a solution to SPECIFICITY, i.e., the problem of targeting underspecified segments, and corresponds to so-called *feature-filling* (or *structure-building*) processes. These two operators are used to define rules where the target is a natural class, the change is a feature specification, and the rule's context for application is optionally specified by natural class-defined triggers to the left and/or right.² This gives rise to the following schemata.

¹In contrast, some theorists assume phonological processes merely tend to be defined by natural classes; for example, Hayes (2009:43) writes that “in most instances, the segments that undergo a rule or appear in the environment of a rule form a natural class”.

²We naïvely assume that targets and triggers are always local to each other; see Dabbous et al. 2024 for a detailed theory of locality and long-distance effects in LP. Furthermore, in lieu of a complete formalization of suprasegmental conditioning of rules, we refer to word boundaries and syllable edges as needed.

(9) Subtraction schema: $[...] \setminus \{...\} / [...] \text{ — } [...]$

(10) Unification schema: $[...] \sqcup \{...\} / [...] \text{ — } [...]$

Following prior suggestions (e.g., Poser 1982, Mester and Itô 1989, Inkelas and Cho 1993, Siptár and Törkenczy 2000), so-called *feature-changing* (or *structure-changing*) processes, such as final devoicing, are thus derived from a subtraction rule followed by a unification rule.³ For example, consider final devoicing in Russian, a feature-changing process.

	NOM.SG.	GEN.SG.	
(11) a.	tsv ^j et	tsv ^j eta	‘color’
b.	prut	pruda	‘pond’

In LP, final devoicing derives from a subtraction rule removing the VOICE specification of final obstruents, followed by a unification rule adding –VOICE. This unification rule can be written quite broadly.

(12) $[-\text{SONORANT}] \setminus \{+\text{VOICE}\} / \text{ — } \%$

(13) $[] \sqcup \{-\text{VOICE}\}$

LP’s ability to generate feature-changing processes via a series of rules means that it does away with the feature-filling/feature-changing rule diacritics used in much prior work (e.g., Kiparsky 1985, Inkelas and Cho 1993, Buckley 1994, Inkelas 1995).

The third operator in LP is the *segment* operator $\mapsto$, which is combined with a null symbol ε to implement insertion and deletion of entire segments.⁴ Schemas are given below.

(14) Insertion schema:

$$\varepsilon \mapsto \begin{array}{c} \text{X} \\ | \\ \{...\} \end{array} / [...] \text{ — } [...]$$

(15) Deletion schema:

$$\begin{array}{c} \text{X} \\ | \\ [...] \end{array} \mapsto \varepsilon / [...] \text{ — } [...]$$

In insertion rules, the X is linked to a feature bundle and a specific segment is inserted. In deletion rules, however, the X is linked to a natural class and any segment matching the natural class is deleted. Because of this, targeting in deletion rules reflects SPECIFICITY. In unification rules, this effect is concealed by vacuous application and unification failure, but there is no such escape hatch for deletion rules.⁵ That is, a unification rule of the form $[\textcircled{\text{I}}] \sqcup \{+\text{HIGH}\}$ has an observable effect only on /I/, whereas a deletion rule targeting $[\textcircled{\text{I}}]$ targets /i, e, I/ both intensionally and extensionally as predicted by DELETE THE RICH.

³This does not exhaust the options available in LP. For example, Gorman and Reiss (2026:§5) decompose Cervara nominal metaphony into a sequence of a unification, a subtraction, and three unifications.

⁴Note that LP does not model segment deletion via subtraction. Subtraction removes features, not segments, and segmental underspecification is representationally distinct from segmental absence.

⁵An anonymous reviewer asks whether one could decompose segment deletion into subtraction followed by deletion of the X. However, DELETE THE RICH entails the second rule—deleting only those X’s linked to a null feature bundle—is unstatable in LP.

2.3 Epistemic boundedness

We now use these tools to reanalyze cases of segment deletion with putative exceptions under the hypothesis that segments which behave differently in the same context are underlyingly distinct. Nearly all phonologists work under the—usually implicit—premise that it is preferable to provide an empirically adequate analysis within the “narrow” phonology rules rather than using various forms of morpheme-specific exceptionality, cophologies, morphophonological rules conditioned on particular lists of morphemes, morphosyntactic rules conditioned on morphosyntactic category or morphemic structure, or suppletion of stems or affixes. This premise is central to the theory of phonology and its interfaces, and it has long been recognized that what we are calling narrow phonology is necessary to account for the characteristic *systematicity* and *productivity* (in the sense of Kenstowicz and Kisseberth 1979:§2) of many sound patterns. Despite this, there is little in the way of mechanistic explanation for how the preference for narrow phonology is expressed during language acquisition. To answer this question we hypothesize that children are *epistemically bounded* (in the sense of Fodor 1980:33f.) to pursue narrow phonological analyses, resorting to these other mechanisms only when the phonological module fails to provide an empirically-adequate solution (see also Gorman and Reiss 2026:§2.4).

LP is a theory of the narrow phonological module, and it follows that the child so endowed with such a module would make use of it when encoding exceptions to deletion. LP permits featural underspecification; DELETE THE RICH provides a mechanism by which segments can be exempted from deletion rules via underspecification. We thus predict the child will make assiduous use of underspecification, as in the following analyses.

3. Hungarian *h*-deletion

Some Hungarian *h*’s delete in the coda, but other *h*’s are realized as coda [x].

(16) Hungarian *h*’s (Siptár and Törkenczy 2000:§8.2.2):

	NOM.SG.	ABL.SG.	NOM.PL.	
a.	tʃɛ	tʃɛ-tø:l	tʃɛfi-ɛk	‘Czech’
	ju	ju-to:l	jufi-ok	‘sheep’
	me:	me:-tø:l	me:fi-ɛk	‘bee’
b.	dox	dox-to:l	dofi-ok	‘musty smell’
	ɛunux	ɛunux-to:l	ɛunufi-ok	‘eunuch’
	me:x	me:x-tø:l	me:fi-ɛk	‘womb’

Siptár and Törkenczy, henceforth S&T, take (16a) to involve exceptional deletion, and derive onset [fi] in (16b), the “systematic pattern”, from underlying /x/.

We claim that in present-day ECH [Educated Colloquial Hungarian] it is the *doh* [(16b)] type that is the systematic pattern... rather than the *cseh* [(16a)] type, which we suggest is not phonological (anymore) and is best considered as suppletive allomorphy. (S&T:275)

In contrast, we hypothesize that roots showing the [fi]–zero alternation end in /h/, and those that show the [fi]–[x] alternation end in /H/, a segment underspecified for CONSONANTAL. A partial feature specification for /h, H/ and their surface correspondents is given below.

(17) Partial feature specifications:

	/h/	[fi]	/H/	[x]
CONSONANTAL	–	–		+
DORSAL				+
VOICE	–	+	–	–

The following deletion rule targets coda /h/.

(18) *h*-deletion:

$$\begin{array}{c} \text{X} \\ | \\ \left[\textcircled{\text{h}} \right] \end{array} \mapsto \varepsilon / \text{ — }]_{\sigma}$$

/H/ is unaffected by this rule. DELETE THE RICH entails that no deletion rule could derive the [h]–zero alternation from underlying /H/ without also deleting coda /h/.

The following rules derive the surface forms of /H/. We assume that /H/ is the only segment not specified for the relevant features at this point of the derivation, so these rules can be stated broadly. The first rule maps /H/ $\rightsquigarrow$ [x] in coda position and the second maps it to [h] elsewhere.

$$(19) \left[\text{ — } \right] \sqcup \left\{ \begin{array}{l} +\text{CONSONANTAL} \\ +\text{DORSAL} \end{array} \right\} / \text{ — }]_{\sigma}$$

$$(20) \left[\text{ — } \right] \sqcup \{ -\text{CONSONANTAL} \}$$

Finally, [h]—whether present underlyingly or derived from /H/ by (20)—is voiced between two vowels or between a sonorant consonant and a vowel.

- (21) a. /tʃɛh-Ek/ $\rightsquigarrow$ [tʃɛfi-ɛk] ‘Czechs’
b. /doH-Ek/ $\rightsquigarrow$ [dofi-ok] ‘musty smells’
c. /kophɔ/ $\rightsquigarrow$ [kopfiɔ] ‘kitchen’

This feature-changing process is implemented by subtraction followed by unification.

$$(22) \left[\textcircled{\text{h}} \right] \setminus \{ -\text{VOICE} \} / \left[+\text{SONORANT} \right] \text{ — } \left[+\text{SONORANT} \right]$$

$$(23) \left[\text{ — } \right] \sqcup \{ +\text{VOICE} \}$$

As stated, the above rules instantiate three critical orderings: (18) must apply before (19), which would counterfactually bleed it; (19) must apply before (20) because the latter is stated more broadly, and (22) must apply before (23) to derive the feature-changing effect. Sample derivations are given below.

		‘Czech’			‘musty smell’		
		NOM.SG.	ABL.SG.	NOM.PL.	NOM.SG.	ABL.SG.	NOM.PL.
UR:		tʃɛh	tʃɛh-tO:l	tʃɛh-Ek	doH	doH-tO:l	doH-Ek
(24)	Harmony:		tʃɛhtø:l	tʃɛhɛk		doHto:l	doHok
	Rule (18):	tʃɛ	tʃɛtø:l				
	Rule (19):				dox	doxto:l	
	Rule (20):						dohok
	Rules (22–23):			tʃɛhɛk			dofiok
SR:		tʃɛ	tʃɛtø:l	tʃɛhɛk	dox	doxto:l	dofiok

The analysis sketched thus far locates the distinct behavior of the two *h*’s fully within the narrow phonology, without recourse to the exceptional suppletion proposed by S&T. Confirmatory evidence for the proposed specification of /H/ comes from its behavior with respect to voice assimilation. Adjacent consonants undergo regressive voice assimilation.⁶ However, /H/ is not a target for voice assimilation, as seen from (25c); e.g., *[me:ɣ-bɛn].

(25) Regressive voice assimilation (S&T:§4.1.1; Péter Siptár, p.c.):

	NOM.SG.	ABL.SG.	INESS.SG.	ALL.SG.	
a.	kəlɔp	kəlɔp-to:l	kəlɔb-bɛn	kəlɔp-hoz	‘hat’
	se:f	se:f-tø:l	se:v-bɛn	se:f-hɛz	‘safe’
	re:s	re:s-tø:l	re:z-bɛn	re:s-hɛz	‘part’
	ʒa:k	ʒa:k-to:l	ʒa:g-bɛn	ʒa:k-hoz	‘sack’
b.	rɔb	rɔp-to:l	rɔb-bɛn	rɔp-hoz	‘captive’
	ka:d	ka:t-to:l	ka:d-bɛn	ka:t-hoz	‘bathtub’
	mɛɛg	mɛɛk-tø:l	mɛɛg-bɛn	mɛɛk-hɛz	‘warmth’
c.	dox	dox-to:l	dox-bɛn	dox-xoz	‘musty smell’
	me:x	me:x-tø:l	me:x-bɛn	me:x-xez	‘womb’

However, /H/ is a trigger of voice assimilation, devoicing preceding obstruents; e.g., /ɔd-Hɔt/ → [ɔt-hɔt] ‘he may give’. This behavior is predicted by the feature specification given above: /H/ is not a target for voice assimilation because it is not +CONSONANTAL, but it is a trigger—i.e., it is *catalytic* in the sense of Gorman and Reiss 2026—of voice assimilation because it is specified for VOICE. Voice assimilation can thus be stated by the following rules (see Reiss 2021:§6.2).

⁶The behavior of *v* poses a related set of difficulties for the analysis of voice assimilation. Reiss (2021, 2024) argues there are also two underlyingly distinct sources for *v*.

$$(26) \left[\begin{array}{c} -\text{SONORANT} \\ +\text{CONSONANTAL} \end{array} \right] \setminus \{ \alpha \text{VOICE} \} / \text{---} \left[\begin{array}{c} -\text{SONORANT} \\ -\alpha \text{VOICE} \end{array} \right]$$

$$(27) [-\text{SONORANT}] \sqcup \{ \alpha \text{VOICE} \} / \text{---} \left[\begin{array}{c} -\text{SONORANT} \\ \alpha \text{VOICE} \end{array} \right]$$

While this seems to support the proposed specification of /H/, a few questions remain; for example, what accounts for the geminates in [dox-xoz, me:x-xez]?

4. Turkish *k*-deletion

Some Turkish *k*'s delete in intervocalic position (all data from Inkelas et al. 2000 unless otherwise noted).

	NOM.SG.	DEF.ACC.SG.	1 SG.POSS.	
(28)	a. konuk	konu-u	konu-um	'guest'
	tyfek	tyfe-i	tyfe-im	'rifle'
	b. sokak	soka-ı	soka-um	'street'
	tykyryk	tykyry-y	tykyry-ym	'spit'
	c. fikir	fikr-i	fikr-im	'idea'
	hykym	hykm-y	hykm-ym	'judgment'
	d. hukuk	hukuk-u	hukuk-um	'law'
	merak	merak-ı	merak-um	'curiosity'
	e. kedi	kedi-ji	kedi-m	'cat'
	kodza	kodza-jı	kodza-m	'husband'
	f. jorum	jorum-u	jorum-um	'comment'
	sanat	sanat-ı	sanat-um	'art'

The most common pattern, affecting hundreds of nouns, is shown in (28a): a root-final *k* preceded by a root vowel deletes after vowel-initial suffixes. Two details should be mentioned. First, some root-final *g*'s also delete in this context (e.g., [katalog]–[katalo-a] 'catalog'), but this only affects a few loanwords, and according to Clements and Sezer (1982:253), speakers tend to pronounce these orthographic root-final *g*'s as [k]. Thus for ease of exposition, the analysis below focuses solely on the behavior of *k*. Secondly, there are a few vowel-initial suffixes which never trigger *k*-deletion; Inkelas and Orgun (1995:768) claim these suffixes all belong to the "Level 2" morphological stratum in which deletion of *k* is expected to be inactive for independent reasons.

In (28b) root-final intervocalic *k* undergoes deletion but root-internal intervocalic *k* is preserved throughout the paradigm. At first glance, this pattern suggests deletion is subject to *non-derived environment blocking* (NDEB). The notion of *derived environment* is usually defined disjunctively: an environment is said to be derived if it is the result either of morpheme concatenation or other phonological processes. (28c) however shows that *k* does not delete in the latter type of derived environment. These roots underlyingly end in two consonants and undergo epenthesis of a harmonic +HIGH vowel in the nom.sg., when this

consonant cluster is word-final (see Inkelas 2009:395f. for details). When the first of the two root-final consonants is *k*, epenthesis renders it intervocalic in the nom.sg., yet the *k* does not delete. Thus deletion of *k* does not conform fully to the canonical NDEB pattern (cf. Inkelas 2000, 2009, 2014:§8.8).

The behavior of vowel-final roots is shown in (28e), suggesting that the harmonic +HIGH vowel in the 1sg. poss. suffix is epenthetic but the featurally-identical vowel in the def. acc. sg. is part of the suffix. Non-dorsal consonants do not delete as shown in (28f).

Although Inkelas is skeptical that *k*-deletion is an instance of NDEB, she also doubts whether the differential behavior of various velars can be handled by the underspecification logic she and colleagues propose for feature-filling processes (e.g., Inkelas and Cho 1993, Kiparsky 1993, Buckley 1994, Inkelas 1995, Inkelas et al. 1997) under the rubric of *inalterability as prespecification*.

Velar Deletion deletes whole segments. One might attempt various strategies for featural prespecification of nonalternating /k, g/ and underspecification of alternating /k, g/, but since the alternation itself is not featural, these attempts would be *far-fetched and un insightful*. (Inkelas 2000:25, emphasis ours)

Yet it is trivial to extend DELETE THE RICH reasoning applied to Hungarian *h*'s to Turkish *k*'s: *k*'s that delete must be more richly specified than *k*'s that do not. For concreteness, we hypothesize that non-deleting [k] is /K/, a segment underspecified for DORSAL, and deleting [k] is the fully specified /k/. The following rules are responsible for the observed patterns. The /-m/ suffix triggers epenthesis of a +HIGH vowel after a word-final consonant.⁷

(29) Epenthesis:

$$\begin{array}{c} \text{X} \\ | \\ \varepsilon \mapsto \left\{ \begin{array}{l} -\text{SYLLABIC} \\ +\text{HIGH} \end{array} \right\} / [+ \text{SYLLABIC}] \text{ — } [-\text{SYLLABIC}] \quad \% \end{array}$$

In (28a,b), the epenthetic vowel also appears after a vowel that becomes root-final due to *k*-deletion, indicating that epenthesis precedes—and is counterbled by—*k*-deletion. This rule does not affect underspecified /K/.

(30) *k*-deletion:

$$\begin{array}{c} \text{X} \\ | \\ [\textcircled{\text{k}}] \mapsto \varepsilon / [+ \text{SYLLABIC}] \text{ — } [+ \text{SYLLABIC}] \end{array}$$

⁷This rule is likely stated too broadly because Turkish permits certain word-final consonant clusters (see Clements and Sezer 1982:§3.5), but the exact generalizations are not settled.

A final unification rule specifies the remaining /K/s as +DORSAL. Other segments are unaffected under the assumption that they are already specified as either +DORSAL or –DORSAL. Alternatively, we can narrow the target class by specifying it as $[\textcircled{\text{K}}]$.

(31) $[\text{ }] \sqcup \{+\text{DORSAL}\}$

Sample derivations are provided below.

		‘street’	‘my street’	‘curiosity’	‘my curiosity’
	UR:	soKak	soKak-m	meraK	meraK-m
(32)	Rule (29) + Harmony:		soKakuum		meraKuum
	Rule (30):		soKauum		
	Rule (31):	sokak	sokauum	merak	merakuum
	SR:	sokak	sokauum	merak	merakuum

Inkelas and Orgun (1995), among others, claim that monosyllables do not undergo *k*-deletion; e.g., [kœk]–[kœkym] ‘(my) root’. They appeal to *ad hoc* notions—BIMORAIC MINIMALITY and FINAL CONSONANT INVISIBILITY—to exempt velars in monosyllables from *k*-deletion. However, there are still some monosyllables which do undergo *k*-deletion.

		NOM.SG.	1 SG.POSS.	
(33)	a.	tʃok	tʃo-um	‘a lot’
		gœk	gœ-ym	‘sky’
	b.	frak	fra-uum	‘tailcoat’
		plak	pla-uum	‘vinyl record’
		brik	bri-im	‘brik (Tunisian pastry)’
				(Kirov et al. 2018)

While these are analytically challenging exceptions for Inkelas and Orgun, they pose no issues for the analysis given here, as shown in the sample derivations below.

		‘my root’	‘my sky’
	UR:	kœK-m	gœk-m
(34)	Rule (29) + Harmony:	kœKym	gœkym
	Rule (30):		gœym
	Rule (31):	kœkym	
	SR:	kœkym	gœym

Under this analysis, /k/ never occurs intervocalically within roots, but this gap is of no great interest: we propose a grammar of Turkish in which a root-internal intervocalic /k/ could never appear on the surface in the first place (cf. Prince and Smolensky 2004:§9.3).

It is thus possible to salvage the underspecification analysis of *k*-deletion for which Inkelas saw little promise. Indeed, the analysis given here demands no new ontological commitments from LP, expressing all the generalizations within the narrow phonology. It is

fully modular in the sense that it makes no reference to morphemic structure as a locus for exceptionality or a condition for application in derived environments.

We assumed above that non-deleting *k*'s are underspecified for DORSAL, but consider an alternative, that non-deleting velars are just prosodically underspecified, e.g., in the sense that they lack an underlying X-slot, but are fully specified, just like velars that delete.

(35) Schematic URs for 'street' and 'law':

- a.

X	X		X	X
s	o	k	a	k

 b.

X	X		X
h	u	k	u

 k

The following schematic derivations then show the operation of Epenthesis, Harmony, *k*-deletion, and a rule or rules associating unassociated segments with X's in lieu of (31).

(36) Epenthesis + Harmony:

- | | | | | | |
|---|---|---|---|---|---|
| X | X | | X | X | X |
| | | | | | |
| s | o | k | a | k | m |

 $\rightsquigarrow$

X	X		X	X	X	X
s	o	k	a	k	u	m

(37) *k*-deletion + default association:

- | | | | | | | |
|---|---|---|---|---|---|---|
| X | X | | X | X | X | X |
| | | | | | | |
| s | o | k | a | k | u | m |

 $\rightsquigarrow$

X	X	X	X	X	X
s	o	k	a	u	m

This approach is consistent with the spirit of DELETE THE RICH—*k*-deletion cannot target a segment not linked to an X-slot—but only up to a point, as default association needs to be able to target unassociated segments. This apparent violation of DELETE THE RICH might be taken to reflect default association's status as an "automatic" convention, applying late in the derivation independent of the featural content, but further theorizing is necessary. In other words, it remains to be seen whether there is any reason to favor prosodic versus featural underspecification for the analysis of Turkish.

5. Conclusions

Under the LP conception of segment deletion rules, the deletion of Hungarian *h*'s and Turkish *k*'s is located entirely within the narrow phonology, without reference to morphemic structure. The resulting phonological rules are exceptionless. DELETE THE RICH follows from LP's minimal ontological commitments rather than stipulations of the theory.

DELETE THE RICH follows directly from the LP formalism, but runs contrary to a common functionalist intuition that less richly specified segments are "weaker" and thus are preferential targets for deletion (e.g., van Oostendorp 2003, Silverman 2011; see Reiss

2025 for discussion). In LP, a substance-free theory, however, this intuition is an example of *substance abuse* (in the sense of Hale and Reiss 2008:§8) and “the mistake of attributing to a mental representation the properties of what it represents” (Pylyshyn 2003:8).

Kiparsky (1993) recognizes that NDEB is a problematic notion and suggests that it would be desirable to reanalyze all putative NDEB effects using underspecification. DELETE THE RICH suggests a means to do so for NDEB effects involving segment deletion, and the LP framework is sufficiently flexible to handle the “exceptional” deletion pattern in Turkish. Of course, it remains to be seen whether all NDEB effects can be similarly reanalyzed.

References

- Bale, Alan, Maxime Papillon, and Charles Reiss. 2014. Targeting underspecified segments: A formal analysis of feature-changing and feature-filling rules. *Lingua* 148:240–253.
- Bale, Alan, and Charles Reiss. 2018. *Phonology: A formal introduction*. Cambridge: MIT Press.
- Bale, Alan, Charles Reiss, and David Ta-Chun Shen. 2020. Sets, rules and natural classes: { } vs. []. *Loquens* 6:e06.
- Blevins, Juliette. 2004. *Evolutionary phonology*. Cambridge: Cambridge University Press.
- Buckley, Eugene. 1994. Prespecification of default features: The two /i/’s of Kashaya. In *Proceedings of the North East Linguistic Society* 24, 17–30.
- Chomsky, Noam, and Morris Halle. 1965. Some controversial questions in phonological theory. *Journal of Linguistics* 1:97–138.
- Clements, G. N., and Engin Sezer. 1982. Vowel and consonant disharmony in Turkish. In *The structure of phonological representations*, ed. by Harry van der Hulst and Norval Smith, 213–254. Dordrecht: Foris.
- Dabbous, Rim, Marjorie Leduc, Charles Reiss, and David Ta-Chun Shen. 2024. Locality is epiphenomenal: Adjacency is opaqueness. In *Proceedings of the 2024 annual conference of the Canadian Linguistic Association*.
- Fodor, Jerry. 1980. Reply to Putnam. In *Language and learning: The debate between Jean Piaget and Noam Chomsky*, ed. by Massimo Piattelli-Palmarini, 325–334. Cambridge: Harvard University Press.
- Gorman, Kyle. 2026. Greek fortition in Logical Phonology. *Canadian Journal of Linguistics* 71. To appear.
- Gorman, Kyle, and Charles Reiss. 2025. How not to acquire exchange processes in Logical Phonology. In *Proceedings of the 2025 annual conference of the Canadian Linguistic Association*.
- Gorman, Kyle, and Charles Reiss. 2026. Metaphony in Substance Free Logical Phonology. *Phonology* 43. To appear.
- Hale, Mark. 2003. Neogrammarian sound change. In *The handbook of historical linguistics*, ed. by Brian D. Joseph and Richard D. Janda, 343–368. Malden, MA: Blackwell.
- Hale, Mark. 2007. *Historical linguistics: Theory and method*. Malden, MA: Blackwell.
- Hale, Mark, and Charles Reiss. 2008. *The phonological enterprise*. Oxford: Oxford University Press.

- Hayes, Bruce. 2009. *Introductory phonology*. Malden, MA: Blackwell.
- Inkelas, Sharon. 1995. The consequences of optimization for underspecification. In *Proceedings of the North East Linguistic Society* 25, 287–302.
- Inkelas, Sharon. 2000. Phonotactic blocking through structural immunity. In *Lexicon in focus*, ed. by Barbara Stiebels and Dieter Wunderlich, 7–40. Berlin: Akademie Verlag.
- Inkelas, Sharon. 2009. Another look at velar deletion in Turkish, with special attention to the derived environment condition. *UC Berkeley Phonology Lab Annual Report* 5:387–403.
- Inkelas, Sharon. 2014. *The interplay of morphology and phonology*. Oxford: Oxford University Press.
- Inkelas, Sharon, and Young-Mee Yu Cho. 1993. Inalterability as prespecification. *Language* 69:529–574.
- Inkelas, Sharon, Aylin Küntay, C. Orhan Orgun, and Ronald Sprouse. 2000. Turkish Electronic Living Lexicon (TELL). *Turkic Languages* 4:253–275.
- Inkelas, Sharon, and Cemil Orhan Orgun. 1995. Level ordering and economy in the lexical phonology of Turkish. *Language* 71:763–793.
- Inkelas, Sharon, Orhan Orgun, and Cheryl Zoll. 1997. The implications of lexical exceptions for the nature of grammar. In *Derivations and constraints in phonology*, ed. by Iggy Roca, 393–418. Oxford: Clarendon Press.
- Kenstowicz, Michael, and Charles Kisseberth. 1979. *Generative phonology: Description and theory*. New York: Academic Press.
- Kiparsky, Paul. 1985. Some consequences of Lexical Phonology. *Phonology Yearbook* 2:85–138.
- Kiparsky, Paul. 1993. Blocking in non-derived environments. In *Studies in Lexical Phonology*, ed. by Sharon Hargus and Ellen Kaisse, 277–313. San Diego: Academic Press.
- Kirov, Christo, Ryan Cotterell, John Sylak-Glassman, Géraldine Walther, Ekaterina Vyloмова, Patrick Xia, Manaal Faruqui, Sabrina J. Mielke, Arya McCarthy, Sandra Kübler, David Yarowsky, Jason Eisner, and Mans Hulden. 2018. UniMorph 2.0: Universal morphology. In *Proceedings of the Eleventh International Conference on Language Resources and Evaluation (LREC 2018)*, 1868–1873.
- Leduc, Marjorie, Charles Reiss, and Veno Volenec. 2024. Votic vowel harmony in Substance Free Logical Phonology. In *The Oxford handbook of vowel harmony*, ed. by Harry van der Hulst and Nancy A. Ritter, 425–436. Oxford: Oxford University Press.
- Lees, Robert B. 1961. *The phonology of Modern Standard Turkish*. Bloomington: Indiana University Publications.
- Lightner, Theodore M. 1971. On Swadesh and Voegelin's 'A Problem in Phonological Alternation'. *International Journal of American Linguistics* 37:227–237.
- Mester, Armin, and Junko Itô. 1989. Feature predictability and underspecification: Palatal prosody in Japanese mimetics. *Language* 65:258–293.
- Ohala, John J. 2003. Phonetics and historical phonology. In *The handbook of historical linguistics*, ed. by Brian D. Joseph and Richard D. Janda, 669–686. Malden, MA: Blackwell.
- van Oostendorp, Marc. 2003. Schwa in phonological theory. In *The second Glot International state-of-the-article book: The latest in linguistics*, ed. by Lisa Cheng and Rint Sybesma, 431–461. Berlin: Mouton de Gruyter.

- Poser, William. 1982. Phonological representation and action at-a-distance. In *The structure of phonological representations*, ed. by Harry van der Hulst and Norval Smith, 121–58. Dordrecht: Foris.
- Prince, Alan, and Paul Smolensky. 2004. *Optimality Theory: Constraint interaction in generative grammar*. Malden, MA: Blackwell.
- Pylyshyn, Zenon W. 2003. *Seeing and visualizing: It's not what you think*. Cambridge: MIT Press.
- Reiss, Charles. 2012. Towards a bottom-up approach to phonological typology. In *Towards a biolinguistic understanding of grammar*, ed. by Anna Maria di Sciullo, 169–191. Philadelphia: John Benjamins.
- Reiss, Charles. 2017. Substance free phonology. In *The Routledge handbook of phonological theory*, ed. by S. J. Hannahs and Anna R.K. Bosch, 425–452. London: Routledge.
- Reiss, Charles. 2021. Towards a complete Logical Phonology model of intrasegmental changes. *Glossa* 6:107.
- Reiss, Charles. 2024. Specificity in rule targets and triggers: Two v's in Hungarian. In *Proceedings of the 2024 annual conference of the Canadian Linguistic Association*.
- Reiss, Charles. 2025. Delete the rich: On the non-existence of 'weak' schwa deletion. Ms., Concordia University.
- Reiss, Charles, and Veno Volenec. 2022. Conquer primal fear: Phonological features are innate and substance free. *Canadian Journal of Linguistics* 67:581–610.
- Shieber, Stuart. 1986. *An introduction to unification-based approaches to grammar*. Stanford: CSLI Publications.
- Silverman, Daniel. 2011. Schwa. In *The Blackwell companion to phonology*, ed. by Marc van Oostendorp, Colin J. Ewen, Beth Hume, and Keren Rice, 628–642. Malden, MA: Wiley-Blackwell.
- Siptár, Péter, and Miklós Törkenczy. 2000. *The phonology of Hungarian*. Oxford: Oxford University Press.
- Volenec, Veno, and Charles Reiss. 2018. Cognitive Phonetics: The transduction of distinctive features at the phonology-phonetics interface. *Biolinguistics* 11:251–294.
- Volenec, Veno, and Charles Reiss. 2019. The intervocalic palatal glide in Cognitive Phonetics. In *Proceedings of the North East Linguistic Society* 49, 255–264.
- Volenec, Veno, and Charles Reiss. 2026. Cognitive Phonetics: The universal phonology-phonetics interface. In *The Cambridge handbook of linguistic interfaces*, ed. by Antonio Fábregas, Laia Mayol, and Yanina Prystauka. Cambridge: Cambridge University Press. To appear.

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